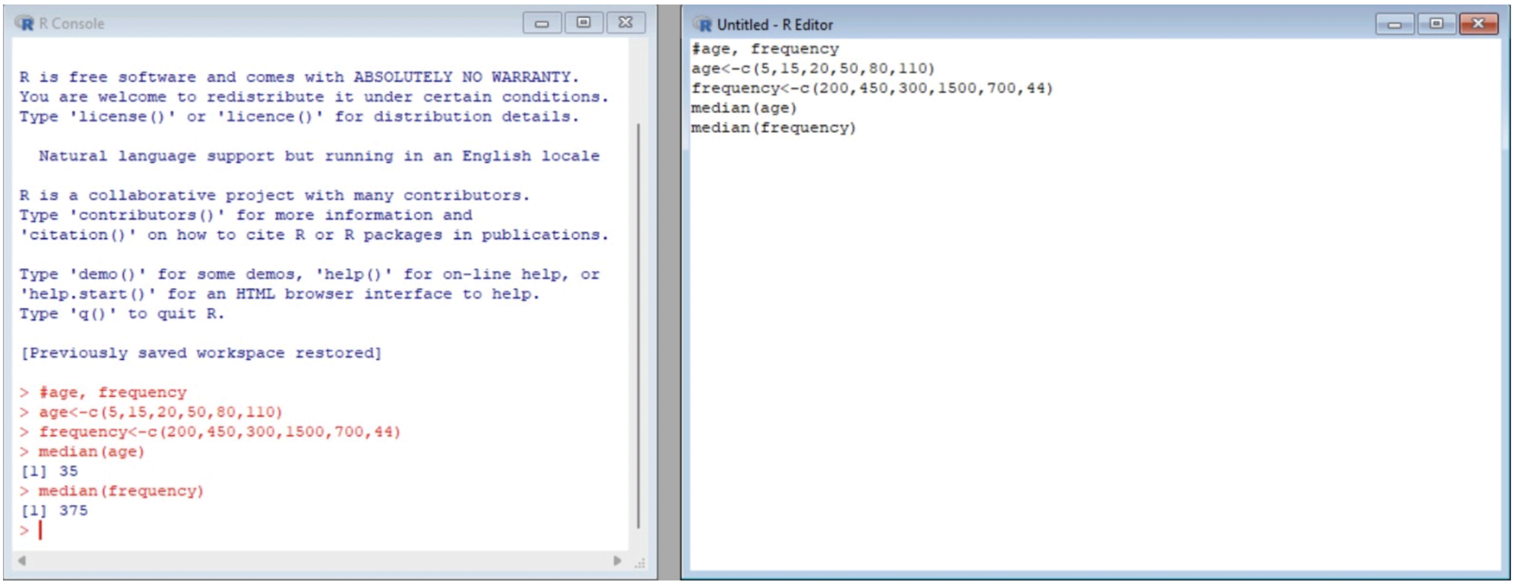
R Console
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Natural language support but running in an English locale
R is a collaborative project with many contributors.
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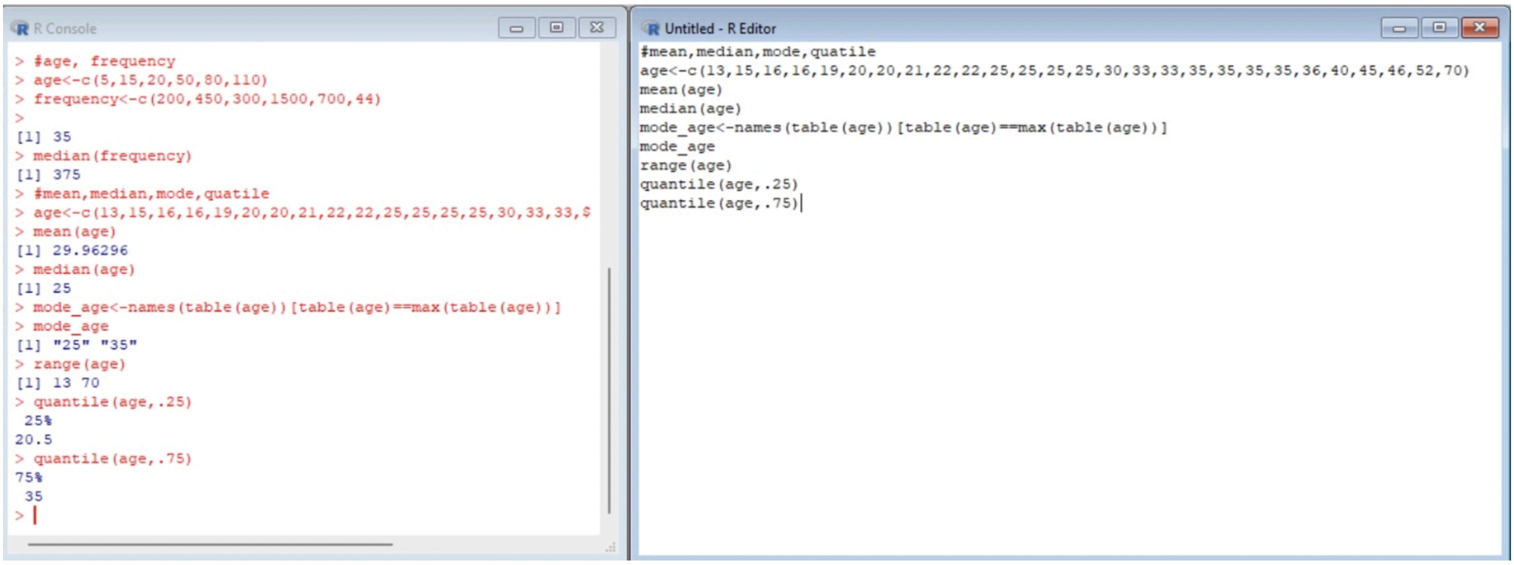
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
[Previously saved workspace restored]
> #age, frequency
> age<-c(5,15,20,50,80,110)
> frequency<-c(200,450,300,1500,700,44)
> median(age)
[1] 35
> median(frequency)
[1] 375
>
Untitled - R Editor
#age, frequency
age<-c(5,15,20,50,80,110)
frequency<-c(200,450,300,1500,700,44)
median(age)
median(frequency)


R Console
> #age, frequency
> age<-c(5,15,20,50,80,110)
> frequency<-c(200,450,300,1500,700,44)
>
[1] 35
> median(frequency)
[1] 375
> #mean,median,mode,quatile
> age<-c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,$
> mean(age)
[1] 29.96296
> median(age)
[1] 25
> mode_age<-names(table(age))[table(age)==max(table(age))]
> mode_age
[1] "25" "35"
> range(age)
[1] 13 70
> quantile(age,.25)
25%
20.5
> quantile(age,.75)
75%
35
>
Untitled - R Editor
#mean,median,mode,quatile
age<-c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70)
mean(age)
median(age)
mode_age<-names(table(age))[table(age)==max(table(age))]
mode_age
range(age)
quantile(age,.25)
quantile(age,.75)

R Console

```
> quantile(age,.75)
75%
 35
> # Data
> x <- c(200, 300, 400, 600, 1000)
> # (a) Min-max normalization (range 0 to 1)
> min_max_norm <- (x - min(x)) / (max(x) - min(x)) min_max_n$
Error: unexpected symbol in "min_max_norm <- (x - min(x)) / $
> # (b) Z-score normalization
> z_score_norm <- (x - mean(x)) / sd(x)
> z_score_norm
[1] -0.9486833 -0.6324555 -0.3162278  0.3162278  1.5811388
> # Data
> x <- c(200, 300, 400, 600, 1000)
>
> # (a) Min-max normalization (range 0 to 1)
> min_max_norm <- (x - min(x)) / (max(x) - min(x))
> min_max_norm
[1] 0.000 0.125 0.250 0.500 1.000
>
> # (b) Z-score normalization
> z_score_norm <- (x - mean(x)) / sd(x)
> z_score_norm
[1] -0.9486833 -0.6324555 -0.3162278  0.3162278  1.5811388
>
```

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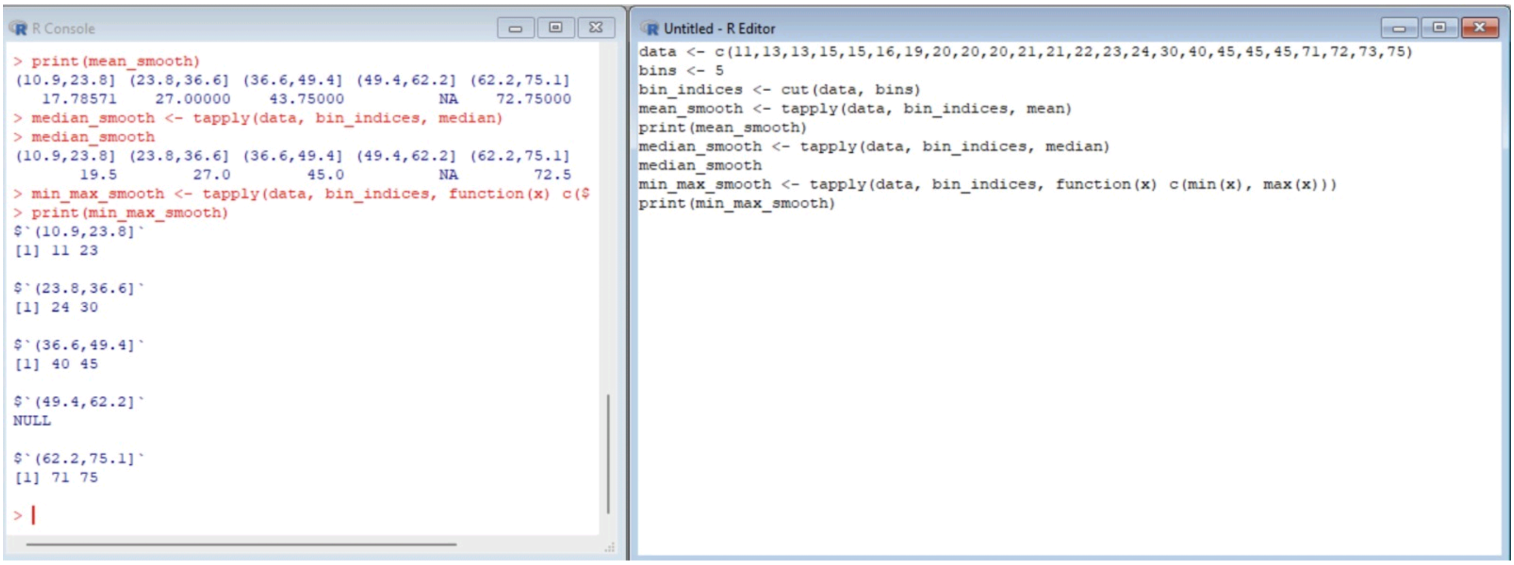
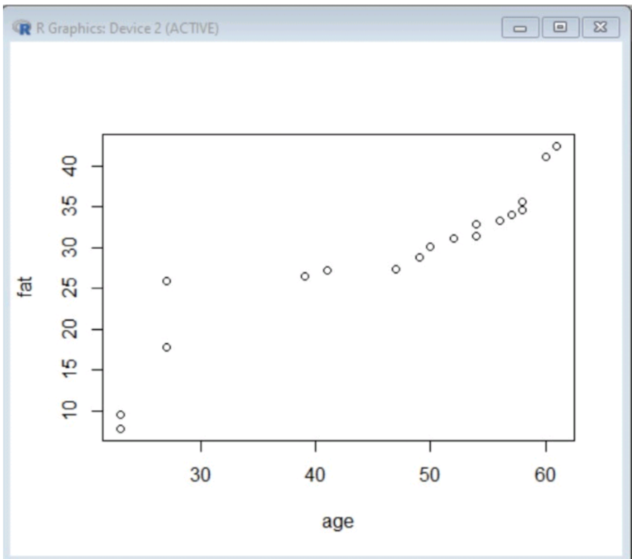
```
# Data
x <- c(200, 300, 400, 600, 1000)

# (a) Min-max normalization (range 0 to 1)
min_max_norm <- (x - min(x)) / (max(x) - min(x))
min_max_norm

# (b) Z-score normalization
z_score_norm <- (x - mean(x)) / sd(x)
z_score_norm
```


R Console
> print(mean_smooth)
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4,62.2] (62.2,75.1]
17.78571 27.00000 43.75000 NA 72.75000
> median_smooth <- tapply(data, bin_indices, median)
> median_smooth
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4,62.2] (62.2,75.1]
19.5 27.0 45.0 NA 72.5
> min_max_smooth <- tapply(data, bin_indices, function(x) c($
> print(min_max_smooth)
$`(10.9,23.8]`
[1] 11 23
$`(23.8,36.6]`
[1] 24 30
$`(36.6,49.4]`
[1] 40 45
$`(49.4,62.2]`
NULL
$`(62.2,75.1]`
[1] 71 75
>
Untitled - R Editor
data <- c(11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,40,45,45,45,71,72,73,75)
bins <- 5
bin_indices <- cut(data, bins)
mean_smooth <- tapply(data, bin_indices, mean)
print(mean_smooth)
median_smooth <- tapply(data, bin_indices, median)
median_smooth
min_max_smooth <- tapply(data, bin_indices, function(x) c(min(x), max(x)))
print(min_max_smooth)



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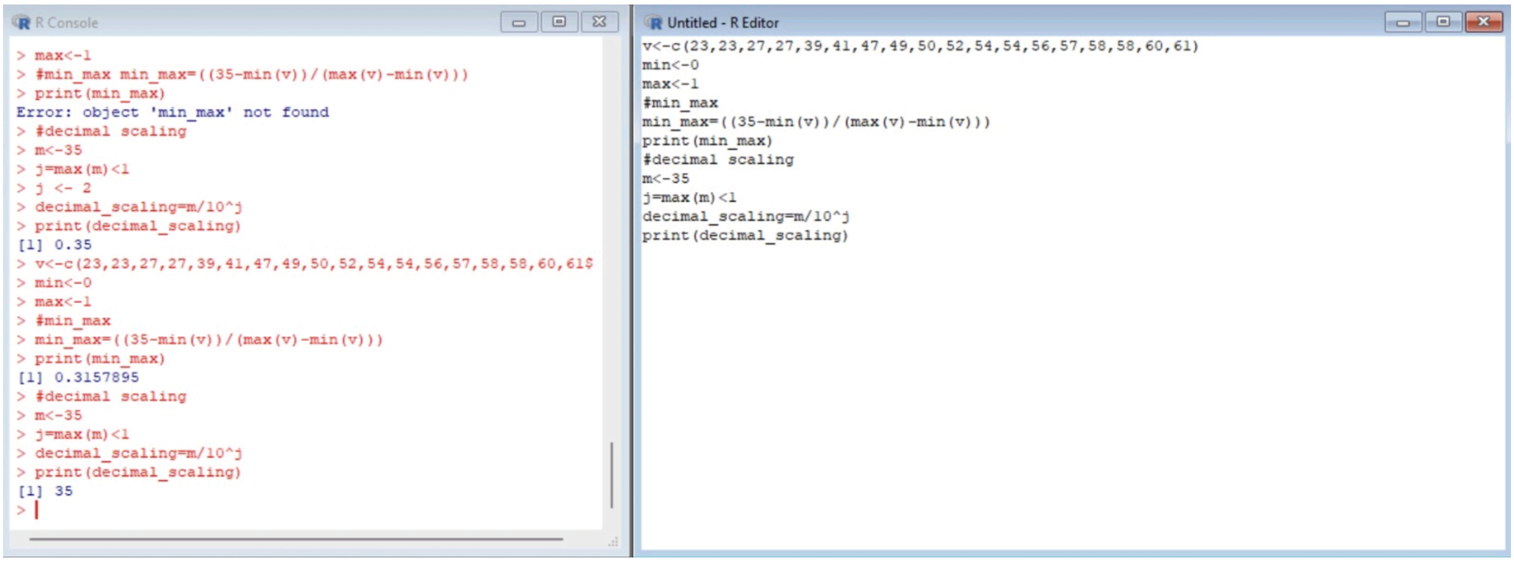
```
age<-c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
fat<-c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,3
mean(age)
median(age)
sd(age)
mean(fat)
median(fat)
sd(fat)
#boxplot
boxplot(age,fat)
#scatter plot scatter.smooth(age,fat)
#qplot
qqplot(age,fat)
```



```
> max<-1
> #min_max min_max=((35-min(v))/(max(v)-min(v)))
> print(min_max)
Error: object 'min_max' not found
> #decimal scaling
> m<-35
> j=max(m)<1
> j <- 2
> decimal_scaling=m/10^j
> print(decimal_scaling)
[1] 0.35
> v<-c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61$
> min<-0
> max<-1
> #min_max
> min_max=((35-min(v))/(max(v)-min(v)))
> print(min_max)
[1] 0.3157895
> #decimal scaling
> m<-35
> j=max(m)<1
> decimal_scaling=m/10^j
> print(decimal_scaling)
[1] 35
>
```

```
v<-c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
min<-0
max<-1
#min_max
min_max=((35-min(v))/(max(v)-min(v)))
print(min_max)
#decimal scaling
m<-35
j=max(m)<1
decimal_scaling=m/10^j
print(decimal_scaling)
```

R Console

```
> decimal_scaling=m/10^j
> print(decimal_scaling)
[1] 0.35
> v<-c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61$
> min<-0
> max<-1
> #min_max
> min_max=((35-min(v))/(max(v)-min(v)))
> print(min_max)
[1] 0.3157895
> #decimal scaling
> m<-35
> j=max(m)<1
> decimal_scaling=m/10^j
> print(decimal_scaling)
[1] 35
> pencils<-c(9,25,23,12,11,6,7,8,9,10)
> mean(pencils)
[1] 12
> median(pencils)
[1] 9.5
> mode=names(table(pencils))[table(pencils)==max(table(penci$
> mode
[1] "9"
>
```

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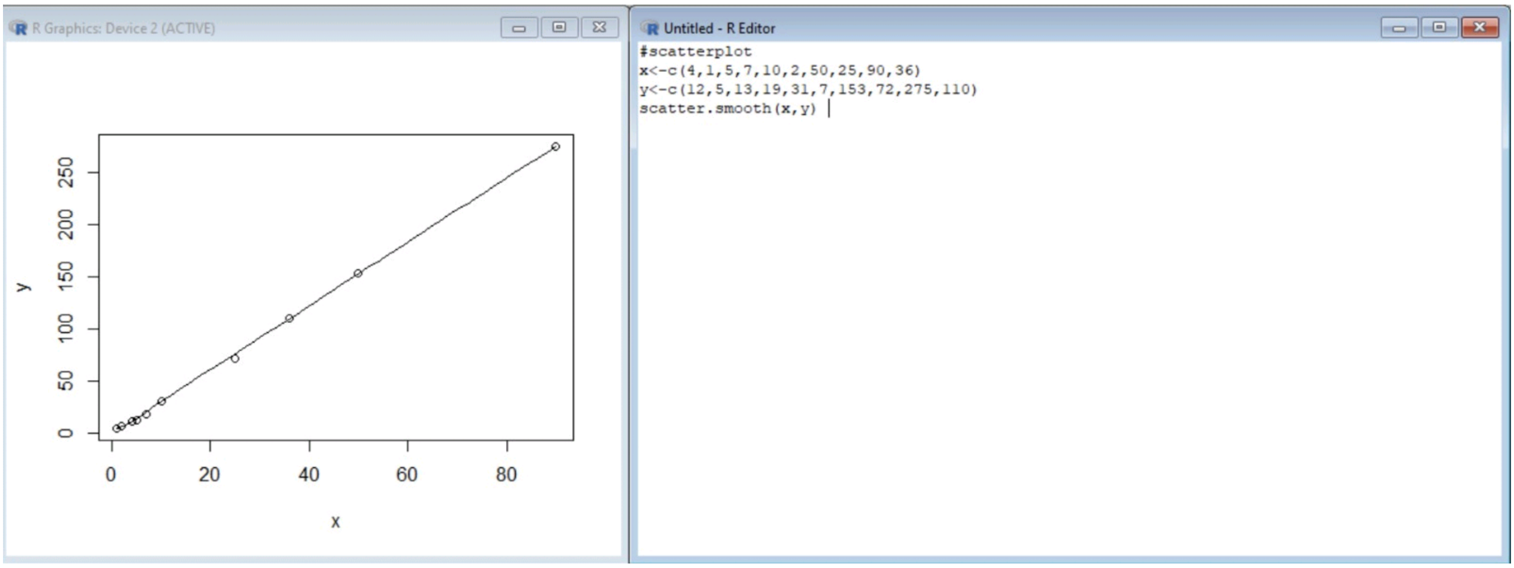
```
pencils<-c(9,25,23,12,11,6,7,8,9,10)
mean(pencils)
median(pencils)
mode=names(table(pencils))[table(pencils)==max(table(pencils))]
mode
```


R Graphics: Device 2 (ACTIVE)
y
x
0
50
100
150
200
250
0
20
40
60
80
Untitled - R Editor
#scatterplot
x<-c(4,1,5,7,10,2,50,25,90,36)
y<-c(12,5,13,19,31,7,153,72,275,110)
scatter.smooth(x,y)

R Console

```
> print("Smoothed data by bin boundaries:")
[1] "Smoothed data by bin boundaries:"
> print(smoothed_data_by_boundaries)
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
          11           30           11           30
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
          11           30           11           30
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
          11           30           11           30
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
          11           30           11           30
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
          11           30           11           30
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
          11           30           11           30
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
          11           30           11           30
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
          11           30           11           30
(32.3,53.7]1 (32.3,53.7]2 (32.3,53.7]1 (32.3,53.7]2
          40           45           40           45
(32.3,53.7]1 (32.3,53.7]2 (32.3,53.7]1 (32.3,53.7]2
          40           45           40           45
(53.7,75.1]1 (53.7,75.1]2 (53.7,75.1]1 (53.7,75.1]2
          71           75           71           75
(53.7,75.1]1 (53.7,75.1]2 (53.7,75.1]1 (53.7,75.1]2
          71           75           71           75
>
```

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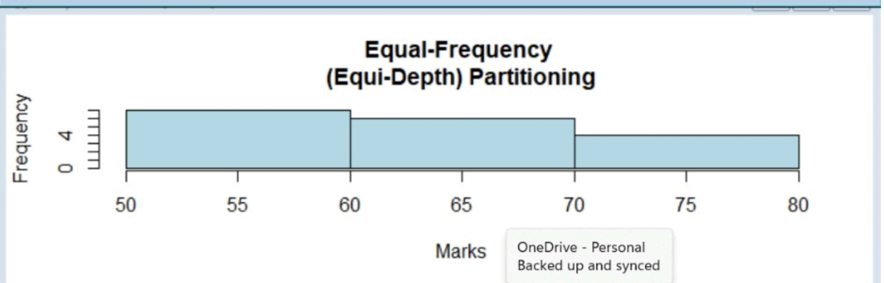
```
marks <- c(55, 60, 71, 63, 55, 65, 50, 55, 58, 59, 61, 63, 65, 67, 71, 72, 75)
num_bins <- 3
bins_eq_frequency <- cut(marks, breaks = num_bins, labels = FALSE)
hist(marks, breaks = num_bins, col = "lightblue", xlab = "Marks", main = "Equal-Frequency
(Equi-Depth) Partitioning")
marks <- c(55, 60, 71, 63, 55, 65, 50, 55, 58, 59, 61, 63, 65, 67, 71, 72, 75)
bin_mean <- tapply(data, cut(data, num_bins), mean)
smoothed_data_by_mean <- unname(bin_mean[as.character(cut(data, num_bins))])
bin_median <- tapply(data, cut(data, num_bins), median)
smoothed_data_by_median <- unname(bin_median[as.character(cut(data, num_bins))])
bin_boundaries <- tapply(data, cut(data, num_bins), function(x) c(min(x), max(x)))
smoothed_data_by_boundaries <- unlist(bin_boundaries[as.character(cut(data, num_bins))])
print("Original data:")
print(data)
print("Smoothed data by bin mean:")
print(smoothed_data_by_mean) print("Smoothed data by bin median:")
print(smoothed_data_by_median)
print("Smoothed data by bin boundaries:")
print(smoothed_data_by_boundaries)
```



R Console

```
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'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

> #IQR,
> SD v<-c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)
Error: unexpected symbol in "SD v"
> IQR(v)
[1] 17.25
> sd(v)
[1] 13.21862
>
```
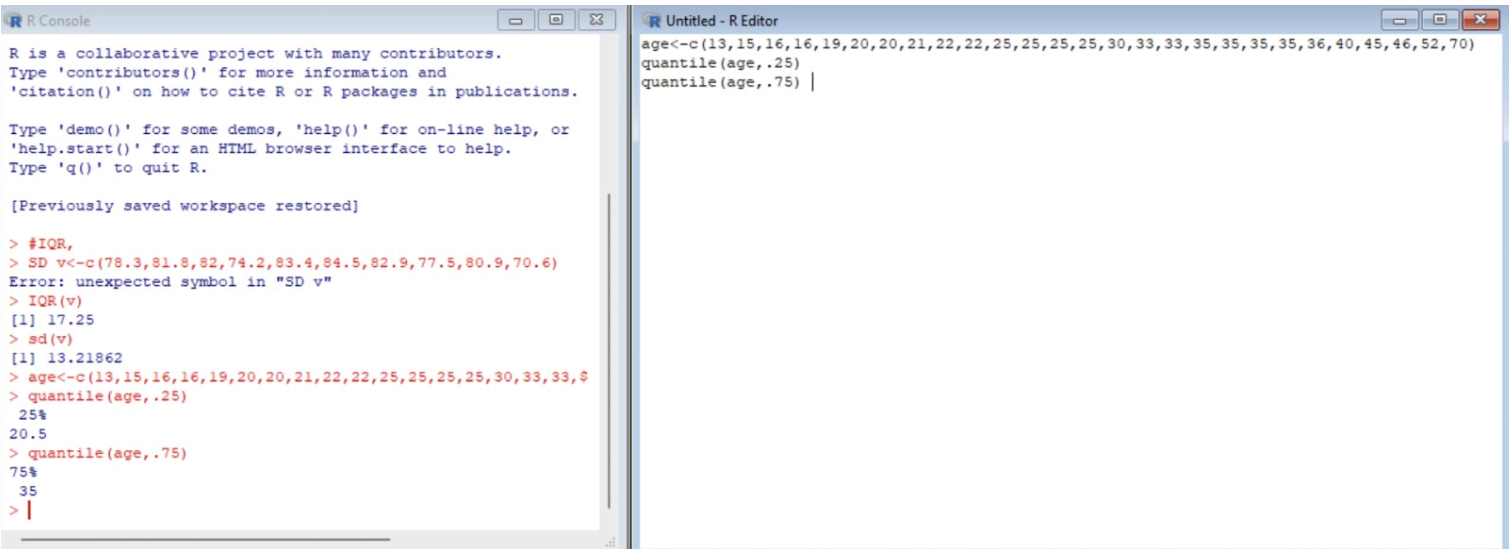

Untitled - R Editor

```
#IQR,
SD v<-c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)
IQR(v)
sd(v)
```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Type 'q()' to quit R.
[Previously saved workspace restored]
> #IQR,
> SD v<-c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)
Error: unexpected symbol in "SD v"
> IQR(v)
[1] 17.25
> sd(v)
[1] 13.21862
> age<-c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,$
> quantile(age,.25)
25%
20.5
> quantile(age,.75)
75%
35
>
Untitled - R Editor
age<-c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70)
quantile(age,.25)
quantile(age,.75)